

FOOD DELIVERY PLATFORM

REQUIREMENTS DOCUMENTATION

Complete Functional & Technical Requirements
for a Full-Featured Food Delivery Platform

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EXECUTIVE SUMMARY

This document outlines comprehensive functional, non-functional, and technical requirements for a full-featured food delivery platform similar to DoorDash and Uber Eats. The platform serves four primary stakeholder groups: Customers, Delivery Partners (drivers), Merchant Restaurants, and Admin/Support teams.

The requirements encompass all aspects of platform design, from user interface requirements to backend infrastructure, security considerations, and compliance obligations.

1. FUNCTIONAL REQUIREMENTS

1.1 CUSTOMER APPLICATION

1.1.1 User Registration & Authentication

- Email/phone number-based sign-up with OTP verification
- Social login integration (Google, Apple, Facebook)
- Secure password management with encryption
- Multi-factor authentication (MFA) option
- Session management with automatic timeout (30 minutes idle)
- Profile management (name, email, phone, profile picture)

1.1.2 Location & Address Management

- Current location detection via GPS
- Address search with auto-complete using maps API
- Save multiple addresses (home, work, favorite locations)
- Set primary/default delivery address
- Edit or delete saved addresses
- Address validation before order placement

1.1.3 Restaurant Discovery & Search

- Browse restaurants by category, cuisine, ratings
- Search restaurants by name, cuisine type, or dishes
- Filter by: delivery time, rating, minimum order, delivery fee, dietary options
- View restaurant details: menu, hours, ratings, reviews, delivery zones
- Restaurant search with geolocation
- Favorite/bookmark restaurants
- View promotions and special offers
- Restaurant scheduling (open/closed status)

1.1.4 Menu Browsing & Item Selection

- View restaurant menu with categories and sub-categories
- Item details: description, price, ratings, ingredients, allergen info

- Customization options: size, add-ons, special instructions
- Item images and nutritional information
- Filter menu items by dietary preferences
- View item availability status
- See recent order history for quick reorder

1.1.5 Shopping Cart Management

- Add items to cart with customizations
- Modify quantities
- Update special instructions for items
- Apply coupon/promo codes
- View itemized cost breakdown
- Estimated delivery time display
- Store cart temporarily (persistent cart across sessions)
- Clear cart functionality

1.1.6 Checkout & Payment

- Multiple payment methods: credit/debit card, digital wallets, cash on delivery
- Saved payment methods
- Secure payment gateway integration (Stripe, PayPal)
- Wallet/credits system with balance tracking
- Apply discount codes and promotional offers
- Tip options (percentage or custom amount)
- Order summary review before confirmation
- Payment error handling and retry mechanism
- Receipt generation and email/SMS confirmation

1.1.7 Order Placement & Management

- Confirm order details and delivery address
- Set delivery time preferences (ASAP or scheduled)
- Special delivery instructions (gate code, buzzer, etc.)
- Order confirmation with order number and estimated delivery time
- Real-time order tracking with status updates
- View order details: items, total, delivery address, contact info
- Order history with filters (active, completed, cancelled)
- Reorder from previous orders with one-click
- Cancel order (with refund logic based on order status)

1.1.8 Real-time Tracking

- Live map view of delivery driver location
- Driver details: name, rating, photo, vehicle info
- Estimated arrival time updates

- Notification when driver is nearby
- Direct communication with driver (call, text, app messaging)
- Route optimization visualization
- Delivery completion confirmation

1.1.9 Ratings & Reviews

- Rate restaurant (1-5 stars with text review)
- Rate delivery experience and driver (1-5 stars)
- Upload photos with reviews
- View aggregate ratings and reviews
- Modify or delete own reviews
- Helpful/unhelpful vote on reviews
- Response to reviews from merchants

1.1.10 Customer Support

- In-app chat support with customer service team
- Reporting system for issues (wrong items, quality, driver behavior)
- Refund request mechanism with proof submission
- Help center with FAQs
- Contact support via multiple channels (chat, email, phone)
- Ticket system with tracking and status updates
- Escalation workflow for critical issues

1.2 MERCHANT (RESTAURANT) APPLICATION

1.2.1 Restaurant Management

- Restaurant profile creation and setup
- Business information: name, cuisine, hours, photos, location
- Bank account and payment setup
- Tax ID and business registration details
- Service area/delivery zone configuration
- Operating hours configuration
- Seasonal closures and holidays
- Pickup vs. delivery availability toggle

1.2.2 Menu Management

- Create, edit, delete menu items
- Organize items into categories and sub-categories
- Item details: name, description, price, images
- Item customization options (size, variants, add-ons)
- Out of stock management for items
- Bulk menu import/export (CSV)
- Version control for menu updates
- Promotional pricing and discounts

1.2.3 Order Management

- Real-time order notifications
- View incoming orders with details
- Accept/reject orders with reason
- Adjust preparation time estimates
- Mark items as ready for pickup
- Integrated kitchen display system (KDS)
- Order status updates (preparing, ready, completed)
- Cancel order with customer notification

1.2.4 Analytics & Reports

- Daily/weekly/monthly order volume
- Revenue analytics with breakdowns
- Popular items and menu performance
- Customer demographics and behavior
- Ratings and feedback analytics
- Operational efficiency metrics (prep time, cancellation rate)
- Delivery performance metrics
- Comparative reports (vs. previous periods)
- Exportable reports (PDF, CSV)

1.3 DELIVERY PARTNER APPLICATION

1.3.1 Partner Registration & Verification

- Phone-based registration
- Background check initiation
- Document upload: ID, license, insurance, vehicle registration
- Verification status tracking
- Approval/rejection notifications
- Profile setup: name, photo, vehicle details

1.3.2 Partner Availability & Status

- Toggle online/offline status
- Current location tracking (background)
- Availability zone configuration
- Working hours logging
- Pause/break functionality
- Logout with daily summary

1.3.3 Delivery Requests & Acceptance

- Real-time delivery request notifications
- Request details: pickup location, delivery location, distance, payment
- Accept/decline delivery requests
- Request expiration and reassignment
- Bulk request view during peak hours
- Navigation to pickup location

1.3.4 Earnings & Finance

- Real-time earnings tracking
- Daily earnings summary
- Weekly/monthly earnings reports
- Bonus opportunities and incentives
- Referral bonuses
- Transaction history
- Payment method setup (bank account, mobile wallet)
- Instant payout options

1.4 ADMIN DASHBOARD

1.4.1 System Overview

- High-level platform metrics (orders, revenue, users)
- Real-time system health monitoring
- Key performance indicators (KPIs)
- Active users, restaurants, delivery partners count
- Platform status and service health

1.4.2 Customer Management

- View all customer accounts
- Search and filter customers
- Customer profile inspection
- Deactivate/ban problematic customers
- Refund management and dispute resolution
- Customer complaint review and resolution

1.4.3 Merchant Management

- Onboard new restaurants
- Approval/rejection of merchant applications
- Restaurant profile verification
- Compliance document review
- Performance monitoring
- Commission/fee configuration per restaurant
- Suspend/ban restaurants

1.4.4 Financial Management

- Revenue tracking and reporting
- Commission calculations
- Partner payouts management
- Refund tracking
- Financial analytics
- Revenue by restaurant/category
- Payment reconciliation

2. NON-FUNCTIONAL REQUIREMENTS

2.1 Performance Requirements

- Page load time: < 2 seconds for 95th percentile
- API response time: < 500ms for 99th percentile
- Real-time tracking update: < 10 seconds
- Concurrent user support: 100,000+ simultaneous users
- Database query optimization for fast data retrieval
- Caching strategy for frequently accessed data
- CDN implementation for static assets

2.2 Scalability Requirements

- Horizontal scaling for backend services
- Auto-scaling based on load
- Database sharding for data distribution
- Message queue implementation for high-traffic handling
- Containerization (Docker/Kubernetes) support
- Load balancer configuration

2.3 Security Requirements

- End-to-end encryption for sensitive data
- HTTPS/TLS for all communications
- PCI-DSS compliance for payment handling
- OAuth 2.0 for authentication
- Role-based access control (RBAC)
- Regular security audits and penetration testing
- Data encryption at rest and in transit
- API rate limiting and DDoS protection
- Secure password hashing (bcrypt, Argon2)
- Two-factor authentication (2FA) for admin accounts
- Regular backup and disaster recovery procedures

2.4 Reliability & Availability

- 99.9% uptime SLA
- Automatic failover mechanisms
- Data replication across multiple regions
- Regular health checks and monitoring
- Graceful degradation during failures
- Incident response procedures
- Transaction rollback capability

2.5 Usability Requirements

- Intuitive UI/UX design
- Accessibility compliance (WCAG 2.1 AA)
- Multi-language support (minimum 10 languages)
- Mobile-first responsive design
- Support for offline functionality where applicable
- Clear error messages and guidance
- Onboarding flows for new users
- Search engine optimization (SEO)

3. SYSTEM ARCHITECTURE

3.1 Technology Stack

- Frontend: React/React Native (web & mobile)
- Backend: Node.js/Python/Go (microservices)
- Database: PostgreSQL (relational), Redis (caching)
- Real-time: WebSocket/Socket.IO
- Message Queue: RabbitMQ/Kafka
- Search: Elasticsearch
- Storage: AWS S3
- Deployment: Kubernetes, Docker

3.2 Microservices Architecture

The platform uses a microservices architecture with the following key services:

- User Service (Authentication & Profile)
- Restaurant Service (Menu & Restaurant Management)
- Order Service (Order Processing & Management)
- Payment Service (Payment Processing & Refunds)
- Delivery Service (Delivery Management & Tracking)
- Notification Service (Alerts, Emails, SMS)
- Rating Service (Reviews & Ratings)
- Search Service (Elasticsearch Integration)
- Analytics Service (Data & Reporting)
- Admin Service (Platform Management)
- Support Service (Customer Support & Ticketing)

3.3 External Integrations

- Google Maps/Apple Maps (directions, geocoding)
- Payment Gateways (Stripe, PayPal, Square)
- SMS Service (Twilio, AWS SNS)
- Email Service (SendGrid, AWS SES)
- Analytics (Google Analytics, Mixpanel)
- Cloud Storage (AWS S3)
- Video KYC (for delivery partner verification)

4. IMPLEMENTATION ROADMAP

4.1 Phase 1 (Months 1-3): MVP Launch

- User authentication & registration
- Basic restaurant search and menu browsing
- Order placement and payment integration
- Basic order tracking
- Customer support system

4.2 Phase 2 (Months 4-6): Core Features

- Real-time driver tracking
- Delivery partner integration
- Rating and review system
- Promotional campaigns
- Admin dashboard basics

4.3 Phase 3 (Months 7-9): Advanced Features

- Analytics and reporting
- Advanced search and filtering
- Merchant analytics
- Loyalty program
- POS system integration

4.4 Phase 4 (Months 10-12): Scale & Optimize

- Performance optimization
- Expanded integrations
- Advanced admin features
- Multi-language support
- Regional expansion

5. SUCCESS METRICS & KPIs

5.1 Customer Metrics

- Daily Active Users (DAU)
- Monthly Active Users (MAU)
- Customer Acquisition Cost (CAC)
- Customer Lifetime Value (CLV)
- Order frequency
- Average order value
- Customer satisfaction score (NPS)
- Repeat customer rate

5.2 Merchant Metrics

- Number of active restaurants
- Average daily orders per restaurant
- Restaurant retention rate
- Commission revenue
- Merchant satisfaction

5.3 Delivery Metrics

- Number of active delivery partners
- Average delivery time
- On-time delivery rate
- Acceptance rate
- Cancellation rate
- Driver satisfaction (rating)
- Driver earnings per delivery

5.4 Platform Metrics

- Total Gross Order Value (GOV)
- Transaction volume
- Platform uptime percentage
- Average app response time
- Customer support response time
- System security incidents

DOCUMENT CONTROL

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